

Math 211 First Exam

12 November 2017

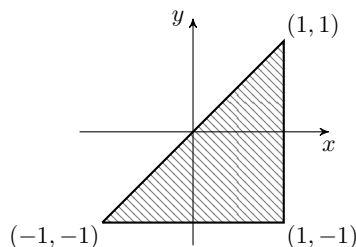
1. [5 pts each] In each part, select the correct answer. No justification is required.

- I. What is the rate of change of the function $f(x, y, z) = xy \sin(xz)$ at the point $(\frac{1}{2}, -2, \pi)$ in the direction of the vector $\langle 2, 2, -1 \rangle$?
- (A) -3 (B) -1 (C) 1 (D) $3/\sqrt{5}$ (E) None of these.
- II. Which of the following is an equation for the tangent plane to the surface $z = xe^{xy^2} + 3y^2$ at the point $(0, 1, 3)$?
- (A) $x + 6y - z = 3$ (C) $6x + y - z = -2$ (E) None of these.
(B) $x + 6y + z = 3$ (D) $x + 6y + z = -3$
- III. Which of the following is an equation for the normal line to the surface $xy + yz + zx = 5$ at the point $(3, -1, 4)$?
- (A) $\frac{x-3}{3} = \frac{y+1}{-7} = \frac{z-4}{2}$ (C) $\frac{x-3}{3} = \frac{y-7}{-1} = \frac{z-2}{4}$ (E) None of these.
(B) $\frac{x-3}{3} = \frac{y+1}{7} = \frac{z-4}{2}$ (D) $\frac{x-3}{5} = \frac{y+1}{5} = \frac{z-4}{5}$
- IV. Which of the following is true about $\lim_{(x,y) \rightarrow (\pi,\pi)} \frac{x \tan^{-1}(y/x)}{x^2 + y^2}$?
- (A) It does not exist by the two-path rule. (D) It is discontinuous at the point $(0, 0)$.
(B) It exists and equals $1/8$. (E) None of these.
(C) It exists and equals 0 .
- V. Which of the following is the set of all points where $f(x, y) = \ln\left(\frac{e^x + e^y}{e^{xy} - 1}\right)$ is continuous?
- (A) $\{(x, y) \mid x > 0 \text{ and } y > 0\}$ (C) $\{(x, y) \mid x < 0 \text{ and } y > 0\}$ (E) None of these.
(B) $\{(x, y) \mid x > 0 \text{ and } y < 0\}$ (D) $\{(x, y) \mid x < 0 \text{ and } y < 0\}$

2. [15 pts] Prove the following:

If f is a differentiable function of x and y , then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a, b \rangle$ and $D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b$.

3. [15 pts] If $z = f(x, y)$ has continuous second-order partial derivatives and $x = r \cos \theta$ and $y = r \sin \theta$, find $\frac{\partial^2 z}{\partial r^2}$.
4. [15 pts] Find the absolute extrema of the function $f(x, y) = e^{x^2 - y^2 - x}$ on the closed triangular region shown.



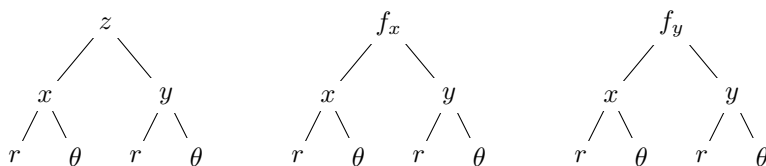
5. [15 pts] Find the curvature of the ellipse $x = 3 \cos t$, $y = 2 \sin t$ at the points $(3, 0)$ and $(0, 2)$.
6. [15 pts] Find an equation for the binormal vector $\mathbf{B}(t)$ at the point $(1, 1, 0)$ for the curve $\mathbf{r}(t) = \langle e^t, e^t \cos t, e^t \sin t \rangle$.

SOLUTIONS

1. I. $\nabla f = \langle y \sin(xz) + xyz \cos(xz), x \sin(xz), x^2 y \cos(xz) \rangle \implies \nabla f(\frac{1}{2}, -2, \pi) = \langle -2, \frac{1}{2}, 0 \rangle$.
On the other hand, the unit vector in the direction of the given vector is $\mathbf{u} = \frac{1}{3}\langle 2, 2, -1 \rangle$.
Therefore, $D_{\mathbf{u}}f(\frac{1}{2}, -2, \pi) = -1$.
- II. $\frac{\partial z}{\partial x} = (1 + xy^2)e^{xy^2} \implies \frac{\partial z}{\partial x}(0, 1) = 1$ and $\frac{\partial z}{\partial y} = 2x^2 y e^{xy^2} + 6y \implies \frac{\partial z}{\partial y}(0, 1) = 6$.
An equation of the tangent plane is $z - 3 = 1(x - 0) + 6(y - 1)$ which simplifies to $x + 6y - z = 3$.
- III. Let $F(x, y, z) = xy + yz + zx$. Then $\nabla F = \langle y + z, x + z, x + y \rangle \implies \nabla F(3, -1, 4) = \langle 3, 7, 2 \rangle$. Hence an equation for the normal line is $\frac{x-3}{3} = \frac{y+1}{7} = \frac{z-4}{2}$.
- IV. The limit is evaluated to $\frac{\pi \tan^{-1}(1)}{2\pi^2} = \frac{1}{8}$ by direct substitution since the function is continuous at (π, π) .
- V. The function f is a composition of continuous functions, hence it is continuous on its domain. Since $e^x + e^y > 0$ for all x , we must have $e^{xy} - 1 > 0$ which holds if and only if $xy > 0$. We conclude that f is continuous on $D_f = \{(x, y) \mid x > 0, y > 0\} \cup \{(x, y) \mid x < 0, y < 0\}$. Hence the correct answer is (E).
2. See pages 948–949 of the textbook.

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = f_x \cos \theta + f_y \sin \theta \implies \frac{\partial^2 z}{\partial r^2} = \frac{\partial}{\partial r}(f_x) \cos \theta + \frac{\partial}{\partial r}(f_y) \sin \theta$$

$$\begin{aligned} 3. \quad &= (f_{xx} \cos \theta + f_{xy} \sin \theta) \cos \theta + (f_{yx} \cos \theta + f_{yy} \sin \theta) \sin \theta \\ &= f_{xx} \cos^2 \theta + 2f_{xy} \sin \theta \cos \theta + f_{yy} \sin^2 \theta. \end{aligned}$$



4. The given triangular region T is a closed and bounded region in $\mathbb{R}^2$. The maximum and minimum values of f are attained at a critical point of f inside T , or at a point on its boundary.

- $f_x = (2x - 1)e^{x^2 - y^2 - x}$ and $f_y = -2ye^{x^2 - y^2 - x}$ are defined everywhere, thus critical points of f are where $f_x = f_y = 0$. Hence $(\frac{1}{2}, 0)$ is the only critical point of f . We have $f(\frac{1}{2}, 0) = e^{-1/4}$.
- On the line segment L_1 from $(-1, -1)$ to $(1, 1)$, we have $y = x$, $-1 \leq x \leq 1$:

$$g_1(x) = f(x, x) = e^{-x}.$$

Since e^x is an increasing function, $g_1(-1) = f(-1, -1) = e^1$ and $g_1(1) = f(1, 1) = e^{-1}$ are the maximum and minimum values of f on L_1 .

- On the line segment L_2 from $(1, -1)$ to $(1, 1)$, we have $x = 1$, $-1 \leq y \leq 1$:

$$g_2(y) = f(1, y) = e^{-y^2}.$$

Similarly to the previous case, $g_2(0) = f(1, 0) = e^0$ and $g_2(-1) = f(1, -1) = g_2(1) = f(1, 1) = e^{-1}$ are the maximum and minimum values of f on L_2 .

- On the line segment L_3 from $(-1, -1)$ to $(1, -1)$, we have $y = -1$, $-1 \leq x \leq 1$:

$$g_3(x) = f(x, -1) = e^{x^2 - x - 1}.$$

Since $g_3'(x) = (2x - 1)e^{x^2 - x - 1}$, Its only critical number is at $x = \frac{1}{2}$. Comparing the values $g_3(-1) = e^1$, $g_3(1) = e^{-1}$ and $g_3(\frac{1}{2}) = e^{-5/4}$ we see that $f(-1, -1) = e^1$ and $f(\frac{1}{2}, -1) = e^{-5/4}$ are the maximum and minimum values of f on L_3 .

- Comparing the obtained values we see that the maximum value of f on T is $f(-1, -1) = e$ and its minimum value if $f(\frac{1}{2}, -1) = e^{-5/4}$.

5. • This ellipse may be represented by the vector function $\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t, 0 \rangle$ with

$$\mathbf{r}'(t) = \langle -3 \sin t, 2 \cos t, 0 \rangle \quad \text{and} \quad \mathbf{r}''(t) = \langle -3 \cos t, -2 \sin t, 0 \rangle.$$

- At the point $(0, 2)$ we have $\cos t = 0$ and $\sin t = 1$, hence $\mathbf{r}' = \langle -3, 0, 0 \rangle$ and $\mathbf{r}'' = \langle 0, -2, 0 \rangle$. Therefore,

$$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3} = \frac{|(-3\mathbf{i}) \times (-2\mathbf{j})|}{|-3\mathbf{i}|^3} = \frac{6}{27} = \frac{2}{9}.$$

6. • We have $\mathbf{r}'(t) = \langle e^t, e^t(\cos t - \sin t), e^t(\sin t + \cos t) \rangle$, thus

$$\begin{aligned} |\mathbf{r}'(t)| &= e^t \sqrt{1 + (\cos t - \sin t)^2 + (\sin t + \cos t)^2} \\ &= e^t \sqrt{1 + (1 - 2 \cos t \sin t) + (1 + 2 \sin t \cos t)} = \sqrt{3} e^t. \end{aligned}$$

This gives $\mathbf{T}(t) = \frac{1}{\sqrt{3}} \langle 1, \cos t - \sin t, \sin t + \cos t \rangle$.

- Now $\mathbf{T}'(t) = \frac{1}{\sqrt{3}} \langle 0, -\sin t - \cos t, \cos t - \sin t \rangle$ and

$$|\mathbf{T}'(t)| = \frac{1}{\sqrt{3}} \sqrt{0 + (-\sin t - \cos t)^2 + (\cos t - \sin t)^2} = \frac{\sqrt{2}}{\sqrt{3}}.$$

This gives $\mathbf{N}(t) = \frac{1}{\sqrt{2}} \langle 0, -\sin t - \cos t, \cos t - \sin t \rangle$.

- At the point $(1, 1, 0)$, we have $e^t = 1$, thus $t = 0$ for which $\mathbf{T}(0) = \frac{1}{\sqrt{3}} \langle 1, 1, 1 \rangle$ and $\mathbf{N}(0) = \frac{1}{\sqrt{2}} \langle 0, -1, 1 \rangle$.
- Finally,

$$\mathbf{B}(0) = \mathbf{T}(0) \times \mathbf{N}(0) = \frac{1}{\sqrt{6}} \langle 1, 1, 1 \rangle \times \langle 0, -1, 1 \rangle = \frac{1}{\sqrt{6}} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \frac{1}{\sqrt{6}} \langle 2, -1, -1 \rangle.$$